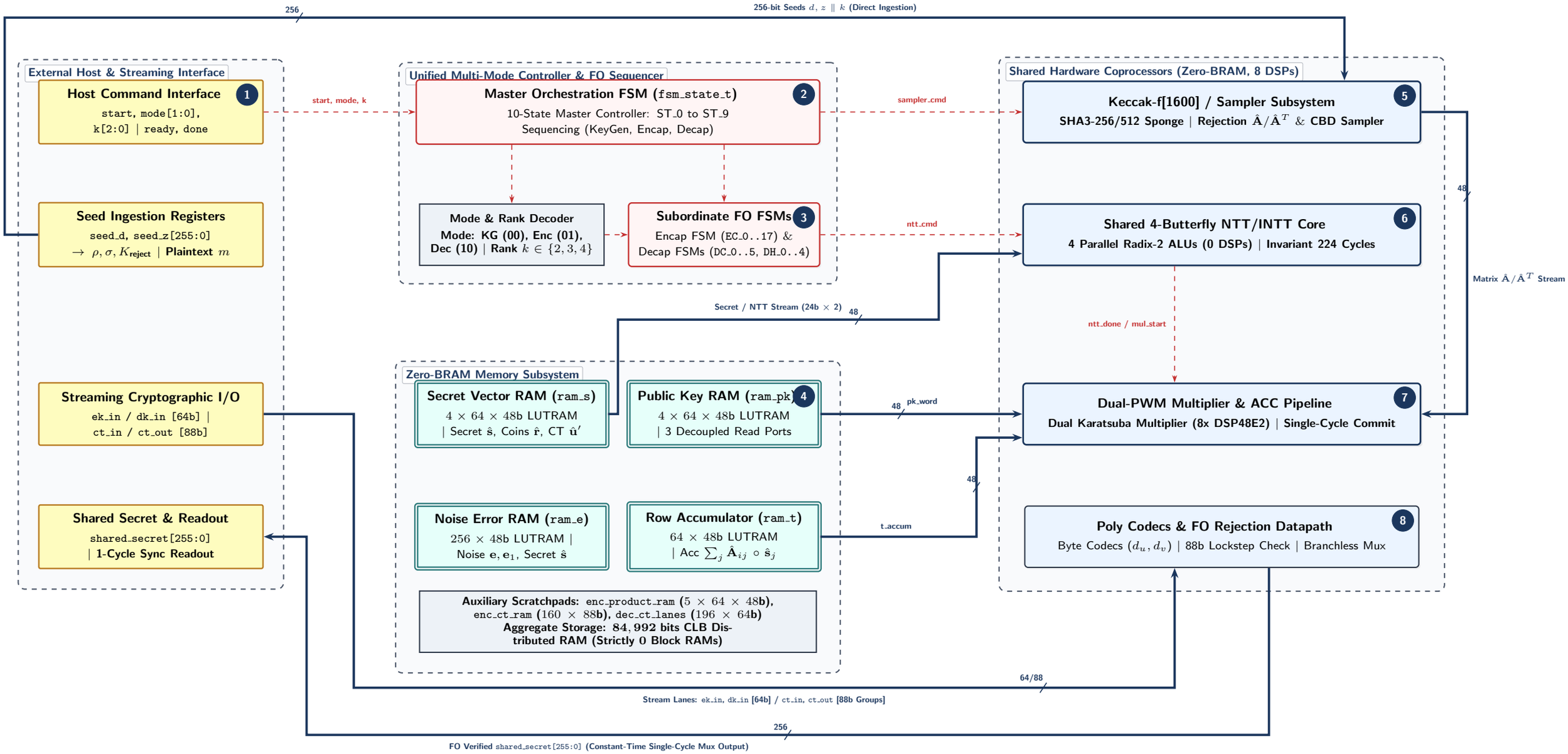
Unified FIPS 203 ML-KEM Architecture: Top-Level System Controller and Datapath

Subsystem 01 Microarchitecture — Unified FIPS 203 KeyGen / Encapsulation / Decapsulation Core — RTL: `mlkem_keygen_top.sv`



ID	Component / Signal	Width / Format	Description
1	Host Command Interface	Control / Handshake	Latches start trigger, operating mode (KeyGen, Encap, Decap), and rank k; exposes status flags and cycle profiler.
2	Master Orchestration FSM	10-State Master FSM	Top-level controller executing states ST_0 through ST_9; manages mode branching and synchronization across all engines.
3	Subordinate FO Controllers	Hierarchical FSMs	Encap datapath FSM (EC_0..17) and Decap FSMs (DC_0..5, DH_0..4) driving PKE encryption, message recovery, and re-encryption.
4	Zero-BRAM Distributed RAM	84,992b LUTRAM	CLB Distributed RAM storing secret vectors, public keys, noise polynomials, and intermediate row sums with strictly 0 BRAMs.
5	Keccak-f[1600] / Sampler	64b Lane Streaming	Shared cryptographic sponge computing SHA3-256/512 digests, SHAKE-128 matrix sampling, and SHAKE-256 CBD error sampling.
6	4-Butterfly NTT/INTT Core	4x Radix-2 ALUs (0 DSP)	Soft-logic transform engine computing forward NTT and inverse INTT in an invariant 224 cycles per polynomial.
7	Dual-PWM Multiplier & ACC	8x DSP48E2 Slices	Pipelined Karatsuba multiplier and 4-stage modular accumulator computing $\hat{\mathbf{A}} \circ \hat{\mathbf{s}} + \hat{\mathbf{e}}$ with single-cycle row commits.
8	Poly Codecs & FO Logic	Codecs + CT Mux	Handles coefficient compression/decompression, lockstep ciphertext comparison, and single-cycle branchless implicit rejection.